

Package insert of Infliximab for Injection Back side

Unfolded size: 805 × 295mm

≥ 18 years of age, including Infliximab in the post-marketing setting. Approximately half the cases were lymphoma and the other cases represented a variety of different malignancies and included rare malignancies usually associated with immunosuppression. A risk for the development of malignancies in children and adolescents treated with TNF-blockers cannot be excluded.

Post-marketing cases of hepatocellular T-cell lymphoma have been reported in patients treated with TNF-blocking agents including infliximab. This rare type of T-cell lymphoma has a very aggressive disease course and is usually fatal. Almost all patients had received treatment with AZA or 6-MP concomitantly with or immediately prior to a TNF-blocker. The vast majority of Infliximab cases have occurred in patients with Crohn's disease or ulcerative colitis and most were reported in adolescent or young adult males. The potential risk with the combination of AZA or 6-MP and Infliximab should be carefully considered. A risk for the development for hepatocellular T-cell lymphoma in patients treated with Infliximab cannot be excluded.

Sodium content

Infliximab contains less than 1 mmol sodium (23 mg) per dose, i.e. essentially "sodium-free". Infliximab is however, diluted in sodium chloride 9 mg/ml (0.9%) solution for infusion. This should be taken into consideration for patients on a controlled sodium diet.

PREGNANCY AND LACTATION

Infant of childbearing potential
Women of childbearing potential should consider the use of adequate contraception to prevent pregnancy and continue its use for at least 6 months after the last Infliximab treatment.

Pregnancy

The moderate number of prospectively collected pregnancies exposed to infliximab resulting in live birth with known outcomes, including approximately 1100 exposed during the first trimester, does not indicate an increase in the rate of malformation in the newborn.

Based on an observational study from Northern Europe, an increased risk (OR, 95% CI, p-value) for C-section (1.50, 1.14-1.96; p = 0.0032), preterm birth (1.48, 1.05-2.09; p = 0.024, small for gestational age (2.79, 1.54-5.04; p = 0.0007), and low birth weight (2.03, 1.41-2.94; p = 0.0002) was observed in women exposed during pregnancy to infliximab (with or without immunomodulators/corticosteroids, 270 pregnancies) as compared to women exposed to immunomodulators and/or corticosteroids only (6,460 pregnancies). The potential contribution of exposure to infliximab and/or the severity of the underlying disease in these outcomes remains unclear.

Due to its inhibition of TNF α , infliximab administered during pregnancy could affect normal immune responses in the newborn. In a developmental toxicity study conducted in mice using an analogous antibody that selectively inhibits the functional activity of mouse TNF α , there was no indication of maternal toxicity, embryotoxicity or teratogenicity.

The available clinical experience is limited. Infliximab should only be used during pregnancy if clearly needed.

Infliximab crosses the placenta and has been detected in the serum of infants up to 12 months following birth. After *in utero* exposure to infliximab, infants may be at increased risk of infection, including serious disseminated infection that can become fatal. Administration of live vaccines (e.g., BCG vaccine) to infants exposed to infliximab *in utero* is not recommended for 12 months after birth. If infant infliximab serum levels are undetectable or infliximab administration was limited to the first trimester of pregnancy, administration of a live vaccine might be considered at an earlier timepoint if there is a clear clinical benefit for the individual infant. Cases of agranulocytosis have also been reported.

Breast-feeding
Limited data from published literature indicate infliximab has been detected at low levels in human milk at concentrations up to 5% of the maternal serum level. Infliximab has also been detected in infant serum after exposure to infliximab via breast milk. While systemic exposure in a breastfed infant is expected to be low because infliximab is largely degraded in the gastrointestinal tract, the administration of live vaccines to a breastfed infant when the mother is receiving infliximab is not recommended unless infant infliximab serum levels are undetectable. Infliximab could be considered for use during breast-feeding.

Drug Interaction
No interaction studies have been performed.

In rheumatoid arthritis, psoriatic arthritis and Crohn's disease patients, there are indications that concomitant use of methotrexate and other immunomodulators reduces the formation of antibodies against infliximab and increases the plasma concentrations of infliximab. However, the results are uncertain due to limitations in the methods used for serum analyses of infliximab and antibodies against infliximab. Corticosteroids do not appear to affect the pharmacokinetics of infliximab to a clinically relevant extent.

The combination of Infliximab with other biological therapeutics used to treat the same conditions as Infliximab, including anakinra and abatacept, is not recommended.

It is recommended that live vaccines not be given concurrently with Infliximab. It is also recommended that live vaccines not be given to infants after *in utero* exposure to infliximab for 12 months following birth. If infant infliximab serum levels are undetectable or infliximab administration was limited to the first trimester of pregnancy, administration of a live vaccine might be considered at an earlier timepoint if there is a clear clinical benefit for the individual infant.

Administration of a live vaccine to a breastfed infant while the mother is receiving infliximab is not recommended unless infant infliximab serum levels are undetectable.

It is recommended that therapeutic infectious agents not be given concurrently with Infliximab.

OVERDOSAGE
Single doses up to 20 mg/kg have been administered without any direct toxic effect. In case of overdose, it is recommended that the patient be monitored for any signs or symptoms of adverse reactions or effects and appropriate symptomatic treatment instituted immediately.

Effects on Ability to Drive and Use Machine
Remiton Infliximab may have a minor influence on the ability to drive and use

machines. Dizziness may occur following administration of Remiton Infliximab.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties
Pharmacotherapeutic group: Immunosuppressants, tumour necrosis factor alpha (TNF α) inhibitors, ATC code: L04AB02.

Mechanism of action
Infliximab is a chimeric human-murine monoclonal antibody that binds with high affinity to both soluble and transmembrane forms of TNF α but not to lymphotxin α (TNF β).

Pharmacodynamic effects
Infliximab inhibits the functional activity of TNF α in a wide variety of *in vitro* bioassays. Infliximab prevented disease in transgenic mice that develop polyarthritis as a result of constitutive expression of human TNF α and when administered after disease onset, it allowed eroded joints to heal. *In vivo*, infliximab rapidly forms stable complexes with human TNF α , a process that parallels the loss of TNF α bioactivity. Elevated concentrations of TNF α have been found in the joints of rheumatoid arthritis patients and correlate with elevated disease activity. In rheumatoid arthritis, treatment with infliximab reduced infiltration of inflammatory cells into inflamed areas of the joint as well as expression of molecules mediating cellular adhesion, chemotaxis and tissue degradation. After infliximab treatment, patients exhibited decreased levels of serum interleukin 6 (IL-6) and C-reactive protein (CRP), and increased haemoglobin levels. Peripheral blood lymphocytes further showed no significant decrease in number or in proliferative responses to *in vitro* mitogenic stimulation when compared with untreated patients' cells. In psoriasis patients, treatment with infliximab resulted in decreases in epidermal inflammation and normalisation of keratinocyte differentiation in psoriatic plaques. In psoriatic arthritis, short term treatment with infliximab reduced the number of T-cells and blood vessels in the inflamed areas.

The ASPIRE study evaluated responses at 54 weeks in 1,004 methotrexate naïve patients with early ($\leq$ 3 years disease duration, median 0.6 years) active rheumatoid arthritis (median swollen and tender joint count of 19 and 31, respectively). All patients received methotrexate (optimised to 20 mg/wk by week 8) and either placebo, 3 mg/kg or 6 mg/kg infliximab at weeks 0, 2, 6 and every 8 weeks thereafter. Results from week 54 are shown in Table 4.

After 54 weeks of treatment, both doses of infliximab + methotrexate resulted in statistically significant greater improvement in the primary endpoint, the total methotrexate alone as measured by the proportion of patients achieving ACRO2, 50 and 70 responses.

In ASPIRE, more than 90% of patients had at least two evaluable X-rays. Reduction in the rate of progression of structural damage was observed at weeks 30 and 54 in the infliximab + methotrexate groups compared to methotrexate alone.

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Effects on ACRs, Structural Joint Damage and Physical Function at week 54, ASPIRE

	Placebo + MTX	Infliximab + MTX
	3 mg/kg	6 mg/kg
Subjects randomised	282	359
Percentage ACR improvement	28.2	35.9
Mean $\pm$ SD ^a	24.8 $\pm$ 59.7	37.3 $\pm$ 52.8
Improvement from baseline in total van der Heijde-modified Sharp score ^b	3.70 $\pm$ 9.61	0.43 $\pm$ 5.82
Median	0.43	0.00
Improvement from baseline in HAQ averaged over time from week 30 to week 54	0.08 $\pm$ 0.63	0.8 $\pm$ 0.65
Mean $\pm$ SD ^a	0.68 $\pm$ 0.63	0.8 $\pm$ 0.65

^a p < 0.001, for each infliximab treatment group vs. control.
^b greater values indicate more joint damage.

^a HAQ = Health Assessment Questionnaire; greater values indicate less disability.
^b p = 0.030 and < 0.001 for the 3 mg/kg and 6 mg/kg treatment groups respectively vs. placebo + MTX.

Data to support dose titration in rheumatoid arthritis came from ATTRACT, ASPIRE and the START study. START was a randomised, multicenter, double-blind, 3-arm, parallel-group safety study. In one of the study arms (group 2, n = 329), patients with an inadequate response were allowed to dose titrate with 1.5 mg/kg increments from 3 up to 9 mg/kg. The majority (67%) of these patients did not require any dose titration. Of the patients who required a dose titration, 80% achieved clinical response and the majority (64%) of these required only one adjustment of 1.5 mg/kg.

Adult Crohn's disease
Induction treatment in moderately to severely active Crohn's disease
The efficacy of a single dose treatment with infliximab was assessed in 108 patients with active Crohn's disease (Crohn's Disease Activity Index (CDAI) $\geq$ 220 $\pm$ 400) in a randomised, double-blind, placebo-controlled, dose-response study. Of these 108 patients, 27 were treated with the recommended dosage of infliximab 5 mg/kg. All patients had experienced an inadequate response to prior conventional therapies. Concurrent use of stable doses of folic acid, oral corticosteroids ($\leq$ 10 mg/day) and/or non-steroidal antiinflammatory drugs (NSAIDs) was permitted.

The primary endpoints were the reduction of signs and symptoms as assessed by the American College of Rheumatology criteria (ACR20) for ATTRACT, landmark ACR-N for ASPIRE, the prevention of structural joint damage, and the improvement in physical function. A reduction in signs and symptoms was defined to be at least a Secondary endpoint (ACR20) in both treated and swollen joint counts, and 3 of the following 5 criteria: (1) evaluator's global assessment, (2) patient's global assessment, (3) functional/disability measure, (4) visual analogue pain scale and (5) erythrocyte sedimentation rate or C-reactive protein. ACR-N uses the same criteria as the ACR20, calculated by taking the lowest percent improvement in swollen joint count, tender joint count, and the median of the remaining 5 components of the ACR response. Structural joint damage (erosions and joint space narrowing) in both hands and feet was measured by the change from baseline in the total van der Heijde-modified Sharp score (0-440). The Health Assessment Questionnaire (HAQ; scale 0-3) was used to measure patients' average change from baseline scores over time, in physical function.

The ATTRACT study evaluated responses at 30, 54 and 102 weeks in a placebo-controlled study of 428 patients with active rheumatoid arthritis despite treatment with methotrexate. Approximately 50% of patients were in functional Class III. Patients received placebo, 3 mg/kg or 10 mg/kg infliximab at weeks 0, 2 and 6, and then every 4 or 8 weeks thereafter. All patients were on stable methotrexate doses (median 15 mg/wk) for 6 months prior to enrolment and were to remain on stable doses throughout the study.

Results from week 54 (ACR20, total van der Heijde-modified Sharp score and HAQ) are shown in Table 3. Higher degrees of clinical response (ACR20 and ACR70) were observed in all infliximab groups at 30 and 54 weeks compared with methotrexate alone.

A reduction in the rate of the progression of structural joint damage (erosions and joint space narrowing) was observed in all infliximab groups at 54 weeks (Table 3).

The effects observed at 54 weeks were maintained through 102 weeks. Due to a number of treatment withdrawals, the magnitude of the effect difference between infliximab and the methotrexate alone group could not be defined.

Table 3
Effects on ACR20, Structural Joint Damage and Physical Function at week 54, ATTRACT

	Control ^a	Infliximab ^b			
	3 mg/kg	3 mg/kg	10 mg/kg	10 mg/kg	All infliximab ^c
Patients with ACR20 response/Patients evaluated (%)	15.88 (17%)	36.86 (42%)	41.86 (48%)	51.87 (59%)	48.81 (52%)
Change from baseline (Mean $\pm$ SD)	7.0 $\pm$ 10.3	1.3 $\pm$ 6.0	1.6 $\pm$ 8.5	0.2 $\pm$ 3.6	0.7 $\pm$ 3.8
Median	4.0	0.5	0.5	0.5	0.5
(Interquartile range)	(0.5, 5.7)	(-1.5, 3.0)	(-2.5, 3.0)	(-1.5, 2.0)	(-1.8, 2.0)

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Patients with no response/Patients evaluated (%)

	13/64 (20%)	34/71 (48%)	35/71 (49%)	37/77 (48%)	44/66 (67%)	150/285 (53%)
HAQ change from baseline over time (patients evaluated)	87	86	85	87	81	339
Mean $\pm$ SD ^a	0.2 $\pm$ 0.3	0.4 $\pm$ 0.3	0.5 $\pm$ 0.4	0.5 $\pm$ 0.5	0.4 $\pm$ 0.4	0.4 $\pm$ 0.4

^a p < 0.001, for each infliximab treatment group vs. control.
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A reduction in the rate of the progression of structural joint damage (erosions and joint space narrowing) was observed in all infliximab groups at 54 weeks (Table 3).

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The co-primary endpoints were the proportion of patients in clinical remission (CDAI $\leq$ 150) at week 30 and time to loss of response through week 54. Corticosteroid tapering was permitted after week 6.

Table 5
Effects on response and remission rate, data from ACCENT 1 (week 2 responders)

	Placebo	Infliximab	p-value
	Maintenance (n = 99)	Maintenance (n = 96)	
Median time to loss of response through week 54	14 weeks	> 40 weeks	< 0.001
Week 54 Clinical remission ^a	23.5	46.2	0.001
Complete fistula response (%) ^b	19.4	36.3	0.009

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